

Code-Layout, Readability and Reusability

Date	2 October 2026
Team ID	SWTID-2026-8368
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S. No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space Python indentation
2	Proper File Structure	Files and folders are logically organized	Yes	app/, templates/, static/, tests/, docs/
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	outline, story, layout, PANELS_DIR
4	Function / Method Names	Functions are descriptively named	Yes	generate_outline, build_comic_layout, save_pdf
5	Code Comments	Inline and block comments explain logic	Partial	Short docstrings and comments on key steps
6	Modular Design	Code is split into reusable functions/modules	Yes	One module per concern
7	No Redundant Code	Duplicate or unused code is removed	Yes	create_comic() shared by both routes
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Pipeline errors return readable HTTP 500

Reusable Components / Modules:

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	create_comic() in routes.py	Python / FastAPI	HTML route and JSON API route	High
2	build_comic_layout (layout_builder.py)	Python	Preview page, PDF export, tests	High
3	generate_image (image_generator.py)	Python / diffusers	Comic pipeline and /test-image	High
4	save_pdf (exporters.py)	Python / fpdf2	PDF export route	Medium

5	Jinja2 templates and CSS	HTML / CSS	Home, preview and export pages	Medium
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Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear separation of routes, AI adapters, layout and export
Readability	4	Short modules, descriptive names, type hints
Reusability	4	Pure layout builder; shared pipeline; configurable models
Documentation / Comments	3	Project documents and docstrings; more inline comments planned
Overall Score	4	Maintainable and tested (24 tests); SDK migration pending